EV Infra Map India

A Charging Infrastructure Decision System for India: Methodology and Implementation Roadmap
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1. Executive summary

A charger map, a gap heat-map and a forecast all answer the question *"what exists?"*. The Ministry of Heavy Industries (MHI), ICAT and state agencies face a harder one: **"Where should the next rupee of public money go, and how will we know it worked?"**

We are building a **Charging Infrastructure Decision System**. The national GIS map is one view of it. The system does four jobs that officials actually have to do:

1. **Trust the data.** Reconcile conflicting official counts (27,737 vs 52,718 stations nationally; 2,400 / 5,765 / 6,805 for Karnataka) into one auditable census. Every record and every hex carries a confidence score, and every district carries an estimate of how many chargers are still missing.
2. **Measure honestly.** Count only supply that is *usable by the vehicles that exist* (working probability, connector compatibility, deliverable power, access hours), not raw plugs.
3. **Separate need from viability.** Show where chargers are needed *and* where no private operator will build them without support. That is where public money adds most.
4. **Make every decision explainable.** Each recommendation comes with a Decision Card: data lineage, confidence, rupees per EV served, and what would change the answer.

Our four headline differentiators, all built for the Grand Finale: **Effective Supply**, the **Trust Layer** (confidence, reconciliation and undercount estimation), the **Market-vs-Mandate viability split with a PM E-DRIVE Allocation Advisor**, and a **policy-versioned norm engine**. The demonstration geography is

Odisha (data-sparse, high need) in depth, contrasted with **Karnataka** (data-rich), on top of a national state-level view.

2. Problem understanding and users

The problem statement names three goals: assess national coverage, identify underserved regions, and align future deployment with actual vehicle populations. We translate each into the question an official asks in a review meeting.

Stakeholder	Real question	What makes them trust and reuse the tool
MHI (PM E-DRIVE owner)	How should the ₹2,000 crore public-charging outlay (about 72,300 chargers) be split across states and cities? Is it working?	Ranked, defensible allocation with rupees per EV served; a sanctioned-vs-installed-vs-effective tracker updated monthly
ICAT / test agencies	Is the method transparent, reproducible and technically sound? Can charger test and certification data plug in?	Open method, versioned runs, validation report, a certified-charger flag
BEE / MoP / state nodal agencies	One trusted charger census. Are the spacing norms met?	Reconciled counts, norm compliance by city and corridor, confidence flags
State transport departments	Which districts and cities need what, by when, under different adoption scenarios?	District need tables, scenario sliders, downloadable site lists
DISCOMs	Where will EV load land, and how much?	kW added per substation area; grid-aware ranking
MoRTH / NHAI	Are highways usable by EV cars, buses and trucks?	Corridor readiness by NH segment; where a vehicle gets stranded
Urban local bodies / MoHUA	Which public land parcels should we offer?	Uncovered cells and ranked candidate parcels

India already has charger locator apps (BEE's EV Yatra portal and operator apps), and a national unified charging app is planned under PM E-DRIVE. We do not build another locator. We build the **planning and decision layer** that sits on top of these sources and is designed to ingest them.

3. Evidence from our pilot (Odisha and Karnataka)

We ran an early version of the pipeline on real VAHAN data for Odisha and Karnataka (Karnataka FY25 EV registrations of 174,003 match the reported 1.73 lakh). Six findings shaped this design:

#	Pilot finding	Design response
1	Official counts conflict by up to 2.8× (Karnataka 2,400 vs 6,805; national 27,737 vs 52,718), mainly because of scope differences (stations vs points, OMC vs all, public vs captive).	Reconciliation waterfall and a single schema with visibility classes (§5).
2	Our open-data census captured about 38% of the official count in Odisha and about 6% in Karnataka. Scaling every district by one completeness factor hides where the missing chargers are.	Multi-source census plus a capture–recapture estimate of the undercount per district (§5.3). Ranges, not points, whenever completeness is below 70%.
3	Observed fast-charger utilisation is 1–5%, against the 4 h/day (~17%) we first assumed for sizing.	Energy-based sizing kept for need; a separate viability layer decides who pays (§7.7, §10).

4	VAHAN-derived stock in Odisha is ~25% below the state figure (1.64 vs 2.19 lakh); unregistered e-rickshaws are missed.	Reconciliation to state figures and an explicit informal e-rickshaw uplift range.
5	By count, Odisha's DC gap is about zero (188 existing vs 115 needed on energy), yet NH-16 has stretches of up to 49 km without a charger.	Gap reported by segment, class and location, not just count; corridor model (§7.6).
6	Public-charging shares were assumptions (2W 10%, e-auto 30%, 4W 20%) and drove the whole need estimate.	Public share derived from home-charging feasibility and treated as the top sensitivity (§7.3).

4. Solution overview

1. Ingest
VAHAN, BEE/MoP, OCM, OSM, operator locators, OMC outlets, scheme lists, Census, WorldPop, OSM roads, substations

2. Census
Normalise → block → match (ML) → cluster → confidence → undercount estimate → reconcile

3. Analytics
Effective Supply, demand and forecast, need sizing, Gap Index, norm compliance, viability, siting

4. Decisions
Decision Cards, Allocation Advisor, scenario and norm versions, uncertainty bands

5. Delivery
National map, dashboard, Allocation Note export, open GeoJSON/OGC layers

Every output is stamped with its **data snapshot date, norm version and scenario**, and every run can be reproduced from its run ID.

5. Data strategy

The problem statement allows open, sample or realistic synthetic data as long as everything is documented. We follow a **data readiness ladder**: T0 open on day one, T1 official public documents, T2 data contracts/MoUs, T3 live operational feeds. Every record carries flags (`is_synthetic`, `is_imputed`, `is_assumed`).

Layer	Sources	Used for	Tier
EV registrations	VAHAN dashboard (RTO × vehicle class × fuel × month); state transport figures; FADA/SIAM retail data as a cross-check	Demand by class, growth, forecast	T0
Official charger counts	BEE / EV Yatra; MoP and MHI statements; Parliament replies; PM E-DRIVE and FAME-II sanctioned lists; state nodal agency lists	Reconciliation target; sanctioned-vs-installed tracking	T1
Open charger data	OpenChargeMap; OpenStreetMap (`amenity=charging_station` with socket tags)	Base census	T0
Operator and OEM networks	Public locators of charge point operators, OEM networks and swap operators, used only where their terms allow; otherwise via data contract	Completeness, connector types, uptime	T2
Fuel retail outlets	OMC retail outlet lists (IOCL, BPCL, HPCL)	Candidate sites with land, power and highway access	T1
Roads and highways	OSM road graph; NHAI network and toll plazas; published traffic counts; PM Gati Shakti layers	Corridor model, drive-time access, range safety	T0/T1

Grid	OSM substations; CEA and state transmission data; SERC EV tariff orders; DISCOM EV connection counts	Grid proxy; load added per area	T0 proxy; T2/T3 actuals
Demographic and economic	Census 2011 with projections; Census/NSS housing tables; WorldPop; GHSL; VIIRS night-lights; open building footprints; OSM POIs	Spatial spread of demand; home-charging feasibility	T0
Contextual (tourism, logistics, land use)	Tourist circuits (e.g. Puri–Konark–Bhubaneswar), logistics hubs, industrial areas, OSM land use, transit and government land	Demand uplift; candidate public land	T0/T1
Public fleets	PM e-Bus Sewa and state transport undertakings' depots	Depot and corridor load	T1/T2
Policy	MoP Guidelines for EV charging infrastructure (17 Sep 2024) and earlier versions; state EV policies	Norm engine; scenario inputs	T0

5.1 Census construction

1. **Ingest** each source into an immutable raw snapshot with a snapshot ID and date.
2. **Normalise** to one schema: site ID; all source IDs; lat/lon; visibility class; operator; port ID; connector; AC/DC; rated kW; supported vehicle classes; hours; status; last-verified date; confidence; flags.
3. **Block** candidate duplicates by geohash-7 cell (~150 m) **and its eight neighbours**, so duplicates that straddle a cell edge are not missed.
4. **Match** candidate pairs with a learned classifier (§8) on distance, operator, connector set, power and name similarity. Names only break ties.
5. **Cluster** matched records (union-find) into sites and keep the most complete attributes.
6. **Score** confidence, estimate the undercount, and **reconcile** to official counts with a published bridge.

Visibility classes. P1 public 24×7; P2 public, restricted hours; S semi-public (mall, hotel, customers only); C captive company/OEM; F fleet depot; H home/housing society. P1 and P2 count fully as public supply; S counts partially; C, F and H are kept as context and to net out demand (an area full of captive depots needs less public supply).

5.2 Confidence score (false positives)

```
Confidence = [1 − Π(1 − r_s) over agreeing source groups] × exp(−age / τ) × attribute completeness
```

r_s is a reliability prior per source (e.g. 0.5 crowd-sourced, 0.8 operator feed, 0.95 field-verified) and τ is a decay constant (12 months). Sources that copy each other (OCM and OSM often share records) are grouped and counted **once**, so copied records cannot inflate confidence. Priors are flagged assumptions, tunable and shown in the Method Card.

5.3 Undercount estimate (false negatives)

Confidence only rates the chargers we have found. The larger risk is the ones we have not found. For each district we estimate total chargers by **capture–recapture** across independent source groups (Lincoln–Petersen, or Chao's estimator when there are more than two groups):

```
N̂ = (n₁ × n₂) / m     n₁, n₂ = chargers found by each source group; m = chargers found by both
```

The estimate is checked against the official count where one exists. The map shows a **"what we don't know" layer**: districts where the estimated missing share is high, so gaps there are reported as ranges

and pushed up the verification queue.

5.4 Reconciliation waterfall and verification queue

A waterfall chart bridges official counts step by step (OMC-only stations → all public stations → points vs stations → captive and semi-public → date differences), turning a disagreement into an auditable explanation. A verification queue ranks sites and districts by *uncertainty × decision impact*, so a state agency verifies the 200 records that matter most rather than 20,000. Results feed back through a Beta-distribution update of each site's working probability.

6. Charger types and compatibility

Supply is only supply for vehicles that can use it. Connector names follow the Indian standards (BIS IS 17017 series) and the MoP guidelines.

Charger / connector	Typical power	AC/DC	Main users	Notes
15 A socket (IEC 60309 industrial / household)	~3.3 kW	AC	e-2W, e-rickshaw, e-3W; some 4W via portable charger	Common at shops, depots and homes
Bharat AC-001	3 × 3.3 kW	AC	2W, 3W, some 4W	Three 15 A outputs; widely used for light EVs
Type 2	7.4–22 kW	AC	4W	Standard public AC for cars; most cars accept 7.2–11 kW
Bharat DC-001 (GB/T)	~15 kW	DC	Early 4W fleet models	Legacy; small installed base
CCS2	30–360 kW	DC	4W, e-buses, e-trucks	Mainstream fast standard
CHAdeMO	~50 kW	DC	A few imported 4W	Legacy tag
LECCS (light-EV combined AC/DC)	Low-power AC/DC	AC/DC	2W, 3W	Light-EV standard; key for 2W/3W
Type 6 (light-EV DC)	Low-power DC	DC	Light EVs	Light-EV fast DC
Battery swap	Pack kWh per swap	n/a	2W, 3W, some 4W	Energy access point: swaps/day × pack kWh

Compatibility weights (draft; versioned data, reviewed yearly):

Segment	Native (1.0)	Partial / adapter (0.3–0.6)	Not usable (0)
e-2W	15 A socket, Bharat AC-001, LECCS, Type 6, swap	Type 2 AC (only with a suitable socket)	CCS2, CHAdeMO, Bharat DC-001
e-rickshaw / e-auto (L3/L5 passenger)	15 A socket, Bharat AC-001, LECCS, swap	Type 6	CCS2, CHAdeMO, Bharat DC-001, Type 2 tethered
3W cargo (L5)	LECCS, Bharat AC-001, 15 A socket	Type 2 (where fitted)	CCS2 (most models), CHAdeMO, Bharat DC-001
4W car	Type 2 AC, CCS2	15 A socket (portable charger), Bharat DC-001 / CHAdeMO (legacy models)	LECCS, Type 6

e-bus	CCS2 (high power), depot charging	GB/T (legacy fleets)	All light-EV and AC slow connectors
e-truck	High-power CCS2	none today	All others

7. Analytical methodology

7.1 Effective Supply (ES)

We replace "number of chargers" with Effective Supply: usable kW for a given vehicle segment at a given location.

```
ES(seg, hex) = Σ over ports p in hex [ min(kW_p, kW_accept(seg)) × P_work(p) × compat(seg, connector_p) × hours_p × access_p ]
```

- **min(kW_p, kW_accept)**: deliverable power. A 7.4 kW-limited car on a 22 kW AC point uses a third of it.
- **P_work**: working probability, starting from a prior by operator and source recency and updated by verification.
- **compat**: the matrix in §6. **hours**: share of the day the port is open. **access**: 1 for public, less than 1 or 0 for semi-public or captive.

Demo moment: toggle "raw count" to "Effective Supply" and watch a district turn from green to red.

7.2 EV stock

Stock by segment = cumulative VAHAN registrations minus segment-specific attrition, reconciled to state transport figures, plus an explicit uplift range for unregistered e-rickshaws. Registration RTO is not where a vehicle is used, so hex-level demand is labelled **modelled** and carries an uncertainty band (§8).

7.3 Public-charging demand (home-charging adjusted)

```
Energy(seg, hex) = stock × km/day × Wh/km
Public share(seg, hex) = 1 − (home- or depot-feasible share × home-charging preference)
```

Home feasibility comes from dwelling type, own parking, apartment share and fleet use (fleets use depots; cars parked at apartments often cannot plug in at home). Proxies: Census/NSS housing tables, building footprints and density indicators, calibrated against any operator session data obtained. This replaces the most arbitrary input in our pilot with something derived from data, and it explains why dense cities need more public chargers per EV than small towns.

7.4 Forecast

- A diffusion model (logistic/Bass) fitted to **monthly** VAHAN series per state and segment, pooled hierarchically so short state series borrow strength from the national curve.
- **Segment-specific saturation levels**, not one cap. Reference points are NITI Aayog's 2030 aspirations (about 80% of 2W/3W sales, 70% of commercial cars, 30% of private cars, 40% of buses). Note that e-rickshaws are already the majority of L3 three-wheelers, so a single 30% ceiling would badly under-forecast 3W.
- Low / Base / High scenarios plus policy-shock cases (subsidy withdrawal or extension). The High scenario is labelled an upper bound where it rests on draft policy.
- Outputs are shown as prediction bands, not just three lines.

7.5 Need sizing

1. **Peak demand:** daily public energy × peak-hour share (8–12%).
2. **Ports by queueing:** an Erlang-C model sizes ports so the probability of waiting stays below a target (e.g. 10%).
3. **Split by class:** light/AC ports for 2W, 3W and slow 4W; DC ports for fast 4W and commercial; depot energy for buses.

7.6 Corridor model

DC need on a corridor = max(norm-spacing sites, flow-based sites), where flow-based need = through-traffic × EV share × share needing a charge × dwell time. We use published NH traffic counts where available, otherwise a clearly flagged gravity-model proxy. Existing DC sites within 1 km of the corridor are netted off before any new site is proposed. A **Highway Range-Safety Index** simulates a typical e-car (real range ~250 km, charging 20–80%) along each NH segment and shows where it gets stranded and the single next site that most improves the corridor (demo: Bhubaneswar → Kolkata on NH-16).

7.7 Gap Index and norm engine

- **Gap = need − Effective Supply**, by segment and charger class, never raw count.
- The Gap Index combines three lenses: norm compliance, demand-to-supply, and drive-time access (OSRM isochrones, % population within 5 / 10 / 20 minutes of a working fast charger). It uses **fixed reference scales** anchored to norms and benchmarks, not min-max across the sample, so a district can be compared over time and with states added later.
- **Policy-versioned norm engine:** norms are stored as data (version, effective date, urban grid, highway spacing, heavy-duty spacing). The active version is the MoP 2024 guidelines (one station per 1 km × 1 km urban grid, every 20 km along highways, every 100 km for long-range and heavy-duty vehicles). The earlier 3 km × 3 km / 25 km version and state-specific norms can be switched in, and compliance is recomputed instantly. The version in use is shown on every screen.

7.8 Viability: Market, Support or Mandate

Every hex and corridor segment with a gap gets one label:

- **Market:** expected utilisation above break-even. Private operators will build; no subsidy needed.
- **Support:** need exists but expected utilisation is below break-even. The support needed is computed site by site as the viability gap (§10.4), not a flat percentage.
- **Mandate:** coverage required by norms or equity even though utilisation will stay very low. These sites need capital *and* operating support, or a mandate on a public operator such as an OMC.

7.9 Siting and the Allocation Advisor

A maximal covering location model with a budget constraint selects sites from candidates (OMC outlets, malls, parking, government and transit land), with a grid-distance penalty. The **PM E-DRIVE Allocation Advisor** ranks picks by **EVs served per rupee of public money** and produces a state-wise allocation within the budget. Subsidy parameters are configurable and default to the scheme's notified rates (including how upstream infrastructure is treated), so the advisor follows the scheme's real design.

> **Decision Card (example).** "Rank 4 because 3,100 uncovered EVs within 5 km; nearest working fast charger 38 km away; substation 1.2 km; label: Support; public money needed ₹X; confidence 0.62. Would drop out of the top 10 if the public share for 4W fell below 15%."

7.10 Validation

- Rolling-origin backtest of the forecast (target MAPE below 20%).
- Leave-one-district-out validation of the demand disaggregation (§8).
- Hold-out siting test: hide 30% of existing chargers and measure how many the model recovers (a sanity check, not an accuracy claim).
- Tornado-chart sensitivity on every assumed parameter, shown in the dashboard.
- The pilot's worked Baleshwar example is kept as a regression test.

8. Where AI is used

We use AI where it beats rules and keep it auditable. Every AI output is labelled as modelled and has a human check.

Task	Method	How we keep it honest
Charger de-duplication across sources	Gradient-boosted classifier on pair features (distance, operator, connector set, power, name similarity), trained on ~500 hand-labelled pairs	Precision/recall reported; uncertain pairs go to the verification queue
Extracting charger lists from official PDFs (Parliament replies, state lists, sanction orders)	LLM-assisted table extraction into the schema	Every extracted row links to its source page; human spot-check of each document before use
Spreading VAHAN RTO counts onto hexes	Gradient-boosted regression on WorldPop, night-lights, building footprints, POIs and road density, trained where DISCOM EV connections or charger session data exist	Leave-one-district-out error published; output labelled "modelled" with bands
Undercount estimation	Capture–recapture across source groups (§5.3)	Checked against official counts where they exist
Forecasting	Hierarchical diffusion model (§7.4)	Rolling backtest; prediction bands

9. Deliverables

PS deliverable	What we build
Functional prototype / web platform	One web application with three personas (national planner, state planner, DISCOM/ULB). Python ETL → PostgreSQL/PostGIS → analytics jobs → FastAPI (REST + GeoJSON) → React + MapLibre front end with pre-computed vector tiles. Governance-ready design: role-based access, audit log, GIGW accessibility, aggregated data only.
Interactive national GIS map	Zoom-dependent aggregation: state → district → H3 hexes → individual sites. H3 resolution 7 (~5 km², ~640,000 cells nationally), with resolution 8 inside built-up areas only. Layers: chargers by class, Effective Supply, Gap Index, norm compliance, highway readiness, EV density, confidence and undercount, substations, candidate sites, PM E-DRIVE sanctioned vs installed. Click any hex for its Decision Card; export the view as GeoJSON/CSV.
Analytical methodology document	Written for ICAT-style technical review, with one Method Card per metric (purpose, formula, inputs, assumptions with ranges, known biases, validation result, version) and an assumptions register.

Charger requirement estimation and recommendation module	For any state, district, city or corridor, scenario and year: points needed by class, gap versus Effective Supply, cost, viability label and ranked sites with Decision Cards.
Decision-support dashboard	Headline indicators: EVs per effective port; drive-time coverage; norm compliance; charging deserts; scenario gap in points and ₹; Market/Support/Mandate split; data confidence and estimated undercount; scheme tracker. National league table, drill-down, corridor view, "what changed since last refresh", scenario and norm sliders, tornado chart, one-click Allocation Note export.
Presentation / demonstration video	Odisha in depth (including NH-16 and the Puri–Konark–Bhubaneswar tourist circuit), contrasted with Karnataka, on the national view.
Scalability roadmap	Section 12: stages with measurable gates, a model-chosen expansion order, costs and cases.

10. Costs, subsidy logic and unit economics

All costs are illustrative planning ranges for showing the method and the order of magnitude. Before the Grand Finale they will be replaced with tender and quotation data and current SERC EV tariffs.

10.1 Cost assumptions (INR, installed)

Item	Low	Central	High	Note
Light point (AC 7.4 kW / LECCS / Type 6), blended	₹0.6 lakh	₹1.0 lakh	₹1.8 lakh	Hardware + installation + metering
DC 30 kW charger	₹8 lakh	₹10 lakh	₹11 lakh	Hardware + installation
DC 60 kW charger	₹13 lakh	₹15 lakh	₹18 lakh	Hardware + installation
Grid connection and transformer per DC site	₹4 lakh	₹8 lakh	₹12 lakh	Strongly site-specific
Civil works, canopy, signage per DC site	₹2 lakh	₹4 lakh	₹5 lakh	Land lease extra
DC highway site (2 × 60 kW), all-in	**₹32 lakh**	**₹42 lakh**	**₹53 lakh**	2 × charger + grid + civil (≈ ₹21 lakh per charger, central)
Annual O&M	~5% of hardware			Plus demand charges as per state tariff

10.2 Worked example: Odisha, FY 2029-30

Pilot outputs: Base need 2,822 light + 115 DC points (energy-based); existing, scaled to the official count, ~553 light + 188 DC; the NH-16 corridor rule gives 44 DC sites (88 points). Light and DC gaps are computed **separately**; a DC surplus is never used to offset a light-charger gap.

Component (Base)	Quantity	Central cost	Range
Light gap (2,822 − 553)	2,269 points	₹22.7 crore	₹13.6–40.8 crore
DC by energy (115 needed vs 188 existing)	0 new by count	₹0	relocation question, not new build
NH-16 corridor DC sites	≤ 44 sites	≤ ₹18.5 crore	₹14.1–23.3 crore
Total capex, Base		**≈ ₹41 crore**	**≈ ₹28–64 crore**

The 44 corridor sites are an upper bound. Each existing DC site already within 1 km of NH-16 removes one new site (₹42 lakh) once the census nets it off.

Scenarios. The pilot's total need is 2,510 / 2,937 / 5,376 points (Low / Base / High). Low and High are split into light and DC using the Base mix (an assumption until the model outputs the split directly). In every scenario the DC energy need (98 / 115 / 211) is covered by existing DC plus the corridor sites (188 + 88 = 276), so the DC requirement is set by the corridor, as the max(energy, corridor) rule intends.

Scenario	Light gap (points)	Light capex	Corridor DC	Total (central)	Range
Low	1,859	₹18.6 crore	₹18.5 crore	≈ ₹37 crore	₹25–57 crore
Base	2,269	₹22.7 crore	₹18.5 crore	≈ ₹41 crore	₹28–64 crore
High	4,612	₹46.1 crore	₹18.5 crore	≈ ₹65 crore	₹42–106 crore

Two observations for MHI: (1) corridor DC cost is **norm-driven** and the same in every scenario, while light-point cost is **adoption-driven** and moves about 2.5× between Low and High; (2) as a sanity check, Odisha's PM E-DRIVE approval of 170 stations / 688 points for ₹40–50 crore is the same order of magnitude, though not like-for-like (it is mostly fast chargers and may be the subsidy portion only).

10.3 Unit economics of a public 60 kW DC charger

Assumptions: all-in capex **₹21 lakh per charger** (half of a 2 × 60 kW site, including grid and civil works); gross margin ₹10/kWh (tariff minus energy cost); fixed costs ₹1.25 lakh/yr (O&M ₹0.75 lakh + demand/connection charges ₹0.5 lakh, which varies widely by state tariff and is waived in some states); 8% cost of capital over 8 years (capital recovery factor 0.174, i.e. ₹3.65 lakh/yr).

Utilisation (share of 24 h at full power)	kWh/day	Gross margin/yr	Net of fixed costs/yr	Simple payback on ₹21 lakh
2% (observed low end)	29	₹1.05 lakh	−₹0.20 lakh	never
5% (observed high end)	72	₹2.63 lakh	₹1.38 lakh	~15 years
10%	144	₹5.26 lakh	₹4.01 lakh	~5.2 years
15%	216	₹7.88 lakh	₹6.63 lakh	~3.2 years

Break-even utilisation including the capital charge: (3.65 + 1.25) lakh ÷ ₹10/kWh ≈ 49,000 kWh/yr ≈ 134 kWh/day ≈ **9.3%**, against observed utilisation of 1–5%. That gap is why public support is needed at all, and the model says where.

10.4 Support needed: the viability gap

Expected utilisation	Annual shortfall	Capital-equivalent support	Share of ₹21 lakh capex	Label
6%	₹1.75 lakh	₹10.1 lakh	~48%	Support
5%	₹2.28 lakh	₹13.1 lakh	~62%	Support
4%	₹2.80 lakh	₹16.1 lakh	~77%	Support
2%	₹3.85 lakh	more than capex	>100%	Mandate: even a full capex grant leaves an operating loss (−₹0.2 lakh/yr)

Coverage norms, not energy demand, drive many DC sites, and those sites will run at low utilisation. So more chargers are needed for coverage than energy alone justifies, and fewer of them are commercially viable. Support is sized to each site's gap; Mandate sites need capital plus operating support (viability-gap funding) or a mandate on a public operator.

10.5 Who pays in Odisha (Base, illustrative)

Illustrative assumptions until the model estimates them site by site: Support sites receive 50% of capex on average (corresponding to ~6% utilisation); Mandate sites receive 100% of capex plus operating support.

Class	Market	Support	Mandate	Public share of capex	Public capex
Light points (₹22.7 crore)	60%	30%	10%	30%×50% + 10%×100% = 25%	₹5.7 crore
NH-16 DC (₹18.5 crore)	20%	30%	50%	30%×50% + 50%×100% = 65%	₹12.0 crore
Total (₹41.2 crore)					**₹17.7 crore ≈ 43%**

Plus operating support for the ~44 Mandate DC chargers, roughly ₹0.2–0.7 lakh per charger per year at 1–2% utilisation (≈ ₹9–32 lakh/yr in total).

The message for MHI: about **42% of the Odisha need (by cost) can be left entirely to the market**, and private capital carries about 57% of total capex. Public money is concentrated on the Support and Mandate sites that private capital will not build, instead of being spread across everything.

10.6 Cost and value of the platform (5 years, illustrative)

Year	Scope	FTE	People	Cloud/data	Field verification	Integration, audit	Total
Y1 (2027)	Pilot: 3–5 states; MoU with MHI/BEE	8	₹1.44 cr	₹0.15 cr	₹0.15 cr (600 sites)	₹0.10 cr	₹1.84 cr
Y2 (2028)	10–15 states, top cities, corridors	14	₹2.80 cr	₹0.35 cr	₹1.25 cr (5,000 sites)	₹0.30 cr	₹4.70 cr
Y3 (2029)	All 36 states/UTs, monthly refresh	20	₹4.40 cr	₹0.60 cr	₹2.70 cr (10,800 sites)	₹0.60 cr	₹8.30 cr
Y4 (2030)	Operate and extend (live feeds)	18	₹4.30 cr	₹0.80 cr	₹2.00 cr (8,000 sites)	₹0.40 cr	₹7.50 cr
Y5 (2031)	Operate and extend	18	₹4.30 cr	₹0.80 cr	₹2.00 cr (8,000 sites)	₹0.40 cr	₹7.50 cr
5-year total							**≈ ₹30 crore**

Loaded cost ₹18–24 lakh per FTE per year; field verification ₹2,500 per site including travel. Hosting on government cloud (MeghRaj/NIC) may reduce cloud costs.

Value for money, on money the platform can actually influence. PM E-DRIVE runs to March 2028 and part of its ₹2,000 crore charging outlay is already sanctioned, so the platform can influence only the uncommitted part of it in 2027–28. Over five years the addressable pool is that remainder **plus** state EV-fund charging budgets, any successor scheme, DISCOM upstream capex, and private CPO investment steered by better siting. The break-even improvement needed is the platform cost divided by that pool:

Addressable 5-year pool (assumption)	₹1,000 crore	₹3,000 crore	₹5,000 crore
Allocation-efficiency gain needed to repay ₹30 crore	3.0%	1.0%	0.6%

Avoiding stranded assets and duplicate sanctions at a single Mandate-class highway site saves ₹42 lakh, so the gains needed are small compared with the money at stake.

Funding and sustainability. (a) A government-funded public good under MHI/BEE, possibly with a small monitoring-and-evaluation share of scheme disbursement; (b) state-level analytics subscriptions; (c) optional paid site-selection and benchmarking reports for CPOs, OEMs and DISCOMs, strictly separated so no commercial user gets privileged access to government data; (d) academic and open-source maintenance through ICAT-linked institutions. The platform never depends on one revenue stream.

11. Implementation roadmap to the Grand Finale

This section is our step-by-step build plan from submission (30 Oct) to the Grand Finale at IIT Bombay (16–18 Dec 2026). Our rule: build a **thin end-to-end slice first** (census → gap → map → one Decision Card for one Odisha district) and deepen it every week. A working demo exists from Week 1, and the front end is never left to the end.

11.1 What the roadmap must cover, and where it is answered

Round 1 requirement	Where it is answered
Data sources to be used	§5 (data layers and tiers) and §11.3 (acquisition plan with fallbacks)
Tools and technologies planned	§11.4 (stack by layer, with reasons)
Development stages	§11.2 (stages with inputs, outputs and exit criteria) and §11.6 (weekly plan and Gantt chart)
How we arrive at the final prototype	§11.5 (build plan for each finale deliverable), §11.8 (testing) and §11.9 (finale readiness)

11.2 Development stages

Stage	When	What we build	Exit criterion (must be true to move on)
D0 Set-up and data pull	31 Oct – 6 Nov (while results are awaited)	Git repository, Docker environment, PostGIS database, schema v1; start monthly VAHAN extraction for Odisha and Karnataka; download OCM, OSM, WorldPop, GHSL, Census and OMC outlet lists; collect official counts and MoP norm text	All T0 sources stored as dated raw snapshots; schema reviewed by the team
D1 Data foundation	Week 1	Ingestion and normalisation for every source; geohash blocking; first matching and clustering; **vertical slice** for one Odisha district (census → gap → map → Decision Card)	Census ≥ 60% of Odisha's official count; the slice runs end to end from one command
D2 Trust and supply	Week 2	ML matching model (500 labelled pairs); confidence score; capture–recapture undercount; reconciliation waterfall; Effective Supply; EV stock and forecast	Matching precision ≥ 0.9 on held-out pairs; forecast backtest MAPE < 20%
D3 Analytics core	Week 3	Demand spread to H3 hexes; need sizing (energy + Erlang-C + corridor); Gap Index on fixed scales; norm engine with versions; viability labels; budget-constrained siting	Need tables for three scenarios; ranked sites with Decision Cards for Odisha; pilot Baleshwar example reproduced
D4 Product	Week 4	FastAPI endpoints; vector tiles; React + MapLibre national map; dashboard and indicators; Allocation Advisor; exports	All five product deliverables working for Odisha in the browser; page loads in under 3 seconds

D5 Second state and documents	Week 5	Karnataka run through the same pipeline; validation report; methodology document with Method Cards; scalability roadmap	Karnataka produced with no code changes, only configuration; documents complete
D6 Finale readiness	12–15 Dec	Demo video, presentation, rehearsals, offline backup, roadmap-evolution change log	Two full rehearsals within 10 minutes; offline demo tested on the presenting laptop

11.3 Data acquisition plan

Data	How we get it	Format	When	Fallback if unavailable
VAHAN EV registrations	Public VAHAN dashboard, monthly by RTO, class and fuel; extracted manually within its terms of use	CSV / Excel	D0–W2	Annual series already pulled in the pilot; state transport figures
Official charger counts	EV Yatra (BEE), MoP/MHI statements, Parliament replies, PM E-DRIVE and FAME-II sanctioned lists	Web pages / PDF	D0–W1	Published national and state totals as reconciliation targets only
Open charger locations	OpenChargeMap API; OSM extract (Geofabrik India) filtered for charging stations	JSON / PBF	D0	None needed (both open)
Operator and OEM locations	Public locators where terms allow; outreach to 3–5 operators for a sample feed	JSON / CSV	W1–W3	Flagged synthetic sample with realistic attributes, documented in the synthetic-data register
OMC fuel outlets	Published outlet lists of IOCL, BPCL and HPCL; OSM fuel stations	CSV	D0–W1	OSM fuel stations only
Roads, highways, routing	OSM road graph (OSRM build for Odisha and Karnataka); NH network	PBF / GeoJSON	D0–W2	Straight-line distance with a road-detour factor (flagged)
Traffic on NH-16	Published NHAI/MoRTH traffic counts	PDF / tables	W2	Gravity-model proxy between corridor cities (flagged)
Population, built-up area, night-lights	WorldPop, GHSL, VIIRS	GeoTIFF	D0	Census 2011 district totals with projections
Housing (home-charging feasibility)	Census and NSS housing tables; open building footprints	CSV / vector	W2–W3	Documented default public shares, with sensitivity range
Grid	OSM substations; SERC EV tariff orders	Vector / PDF	W2	Distance to nearest substation as a proxy
Boundaries	Open state and district boundary datasets	GeoJSON	D0	None needed

Every dataset is logged in a data register (source, licence, date, snapshot ID, synthetic or not), which feeds the methodology document and the rule 5 / rule 6 disclosures.

11.4 Tools and technologies

Layer	Tools	Why
Data processing	Python, pandas, GeoPandas, Shapely, rasterio, H3	Standard, free, fast to develop; H3 gives consistent hexes nationally

Storage	PostgreSQL + PostGIS	Spatial queries, snapshots and run tables in one place
Modelling	scikit-learn, LightGBM, statsmodels, PuLP / OR-Tools (siting), custom Erlang-C	Explainable models; open-source optimisation solvers
Routing	OSRM on OSM	Free drive-time isochrones and corridor simulation
AI-assisted extraction	LLM-assisted PDF table extraction with source-page links	Turns official PDFs into rows quickly; every row is human-checked
API	FastAPI (REST + GeoJSON)	Fast, typed, auto-documented
Map and dashboard	React, MapLibre GL, ECharts, PMTiles vector tiles	Open-source, smooth at national scale; no paid map licences
Fallback front end	Streamlit + Folium	Used only if the React front end slips
Engineering	Git and GitHub, Docker Compose, pytest, GitHub Actions	Reproducible runs, automated tests, same set-up on every laptop
Hosting for the demo	One cloud VM (or free tier) plus a full local copy on the presenting laptop	The demo works even without venue internet

11.5 Build plan for each finale deliverable

Finale deliverable (PS)	Components we build	Built in	Done when
Functional prototype / web platform with a unified view	Census pipeline, PostGIS, API, web app with three personas	W1 slice → W4 complete	One URL shows the reconciled charger census for India, Odisha and Karnataka
Interactive national GIS map	Vector tiles for state, district, hex and site levels; layers for chargers, Effective Supply, gaps, norms, corridors, confidence and undercount	W1 basic → W4 full	Zoom from India to a single site; click a hex for its Decision Card
Analytical methodology document	Method Cards per metric, assumptions register, validation report	Written alongside each stage; finalised W5	Every number on screen traces to a Method Card
Charger requirement estimation and recommendation module	Forecast, need sizing, corridor model, viability, siting, Allocation Advisor	W2–W4	For any district, corridor or state: points needed by class, cost, viability label and ranked sites
Decision-support dashboard	Indicator panel, league table, scenario and norm switches, tornado chart, Allocation Note export	W4	A non-technical user finds the top-priority district in under 2 minutes
Presentation / demonstration video	Five-minute demo story recorded on Odisha (NH-16, Puri–Konark–Bhubaneswar) with Karnataka contrast	12–15 Dec	Video under 6 minutes, subtitled, uploaded and also stored offline
Scalability roadmap	Section 12, updated with real run times and costs from the prototype	W5	Gates and costs backed by measured numbers

11.6 Weekly plan

Week	Focus	Output
D0 (31 Oct – 6 Nov)	Set-up; data pull; schema	Raw snapshots; schema v1
W1 (7–13 Nov)	Ingestion; dedupe; **vertical slice end to end**; label 500 match pairs	Working skeleton; first census
W2 (14–20 Nov)	Matching model; confidence; undercount; waterfall; Effective Supply; forecast	Reconciliation chart; three-scenario stock forecast
W3 (21–27 Nov)	Demand spread; need sizing; Gap Index and norm engine; viability; siting; validation	Ranked sites; validation report draft
W4 (28 Nov – 4 Dec)	API; map; dashboard; Allocation Advisor; Decision Cards; national state-level layer	Feature-complete platform for Odisha
W5 (5–11 Dec)	Karnataka run; methodology document; scalability roadmap; bug fixing	Two-state platform and documents
W6 (12–15 Dec)	Demo video; presentation; rehearsals; offline backup	Video, deck, tested offline demo

Workstream	D0	W1	W2	W3	W4	W5	W6
Data pull and census	■	■	■			▒	
Trust layer (matching, confidence, undercount)		▒	■	▒			
Forecast and need sizing			■	■		▒	
Gap, norms, viability, siting		▒		■	▒		
API, map, dashboard		▒	▒	▒	■	▒	
Second state (Karnataka)	▒					■	
Validation and documents			▒	▒	▒	■	
Video, deck, rehearsal						▒	■

Dark cells: main focus that week. Light cells: lighter work (data download, early prototype, documentation alongside the build).

11.7 Scope priorities

Priority	Items
Must	Census with confidence, undercount estimate and reconciliation waterfall; Effective Supply; norm engine with version switch; Gap Index on fixed scales; need module (energy + queue + corridor); **viability layer and Allocation Advisor**; siting with Decision Cards; national state-level map plus Odisha in depth; dashboard with scenarios
Should	Karnataka as the second state; drive-time access; tornado chart; ML demand disaggregation (with population-weight fallback)
Could	Range-Safety Index for NH-16; home-charging-adjusted public share (with documented default shares as fallback); mobile verification form
Roadmap only	Live OCPI feeds, DISCOM capacity data, national refresh automation

Member	Roles	Responsibilities

Shrimon Mishra (Team Leader)	Modelling lead; overall coordination	Matching model, EV stock and forecast, need sizing (energy, Erlang-C, corridor), Gap Index, viability, siting and Allocation Advisor, validation suite; overall plan and integration; compiles the methodology document; presents at the finale
Shinjan Sarkar	Data lead; GIS and back end	Data acquisition and VAHAN extraction, schema, ingestion, census and reconciliation; PostGIS, H3, OSRM routing, vector tiles, FastAPI; deployment and offline build
Khushi Jha	Front end and dashboard; demo	React + MapLibre national map, dashboard, Decision Cards, exports; user testing; demo video

Each owner writes the Method Cards for their own components. Shrimon reviews every model output before it reaches the dashboard, and Shinjan and Khushi pair on the API–front end contract in Week 1.

Working rhythm: a 15-minute sync twice a week, a shared task board, and a demo of the current build to the whole team every Sunday.

11.8 Testing and quality assurance

- **Data tests** on every run: row counts, coordinates inside India and inside the right district, duplicate rate, missing-attribute rate.
- **Model tests:** matching precision/recall on held-out pairs; forecast rolling backtest; leave-one-district-out demand error; hold-out siting recovery.
- **Regression tests:** the pilot's Baleshwar example and the Odisha cost tables (§10) must reproduce exactly.
- **Reproducibility:** the same snapshot and run ID give identical outputs on two laptops.
- **User test:** two people outside the team each try the dashboard with three tasks ("find the worst district", "explain this site", "change the scenario"); we fix whatever slows them down.

11.9 Finale readiness

- **Three ways to demo:** live on the hosted version; the same build running offline on the presenting laptop (Docker, cached tiles); and the recorded video if both fail.
- **Roadmap-evolution log:** we keep a dated change log from this Round 1 plan, so the walkthrough the finale requires ("how the solution evolved from the Round 1 roadmap") is ready and honest.
- **Q&A preparation:** a one-page answer sheet for the questions we expect (data gaps, utilisation assumptions, why Odisha, how it scales, AI use), each pointing to a Method Card.

> **Five-minute demo story.** (1) Two official numbers disagree: show the waterfall and the undercount layer. (2) Switch raw count to Effective Supply and watch a district turn red. (3) Switch the norm version and the adoption scenario. (4) Open a Decision Card and show ₹ per EV served. (5) Show the Market/Support/Mandate split, then drive an e-car along NH-16 and show where it gets stranded.

11.10 Schedule risks and fallbacks

If this slips…	…we do this instead
Monthly VAHAN extraction takes too long	Use the annual series already pulled in the pilot; add monthly data for Odisha only
Matching model underperforms	Rule-based matching on distance + operator + connector (pilot method), with uncertain pairs flagged

ML demand spread is not better than the simple method	Population and built-up-area weights, labelled "modelled"
React front end is late	Streamlit + Folium front end on the same API
Karnataka run is incomplete	Show Karnataka at district level only; Odisha stays in full depth
Operator data never arrives	Documented synthetic sample, clearly flagged, so the method still demonstrates
Venue internet fails	Offline laptop build, then recorded video

12. Scalability roadmap (5 years)

The roadmap is data-driven in two ways: each stage is gated by metrics the platform computes, and the order of expansion is chosen by a score the platform produces.

Stage	Timing	Scope	Gate (measured by the platform)
S0 Prototype	Dec 2026	Odisha in depth, Karnataka for contrast, national state-level view; all deliverables in demo form	Census ≥ 60% of Odisha's official count; VAHAN gap explained; forecast backtest MAPE < 20%
S1 Sponsor pilot	2027 (Y1)	3–5 states; MoU with MHI/BEE for official lists and a bulk VAHAN feed; first Allocation Note	Census completeness ≥ 70%; > 50% of supply with confidence ≥ 0.7; at least one state nodal agency uses the Allocation Note
S2 Multi-state	2028 (Y2)	10–15 states, top ~50 cities, major NH corridors, e-bus depots; CPO/OEM data contract live; PM E-DRIVE tracker before the scheme ends in March 2028	Census ≥ 80%; ≥ 10 CPO/OEM feeds; ≥ 70% of top-impact records verified; forecast error stable
S3 National	2029 (Y3)	All 36 states/UTs; monthly refresh; Gati Shakti layers; DISCOM load module	Census ≥ 85%; data freshness ≤ 35 days; ≥ 20 states using outputs
S4 Live and extended	2030–31 (Y4–5)	Live OCPI-style feeds; heavy-duty and e-truck corridors; megawatt-class charging; swap parity	Live uptime for ≥ 40% of fast chargers; norm compliance reported quarterly

Expansion order chosen by the model:

```
Rollout priority(state) = 0.4 × Need (gap × EV growth) + 0.3 × Data readiness + 0.3 × Institutional readiness
```

Weights are tunable and shown. Each wave mixes a high-need, data-sparse state (Odisha-like) with a data-rich one (Karnataka-like), so the platform is tested at both ends.

Interoperability. All outputs are published as open-standard layers (GeoJSON, OGC services) ready to overlay in PM Gati Shakti and state GIS platforms. A short OCPI-lite data contract lets CPOs and OEM networks submit data in return for benchmark reports and subsidy-eligibility evidence. Charger depots, swap stations and plug-in chargers share one energy-access schema with kWh-equivalent capacity.

12.1 Planning cases

Dimension	Worst case	Base case	Best case
Data access	No bulk VAHAN feed; no MoU; CPOs decline	MoU for official lists and monthly VAHAN extract; ~one-third of CPOs share	Data sharing becomes a subsidy condition; live feeds; DISCOM capacity data

Adoption (Odisha FY30 stock)	6.8 lakh	8.3 lakh	14.4 lakh (needs a notified high-adoption policy)
Charger need (Odisha FY30 points)	2,510	2,937	5,376
States covered (Y1 / Y3 / Y5)	3 / 8 / 15	5 / 36 / 36	8 / 36 / 36
Census completeness (Y1 / Y3 / Y5)	50% / 60% / 65%	70% / 85% / 90%	85% / 93% / 96%
Annual run cost (steady state)	~₹1.2 crore (5 FTE, open data only)	~₹7.5 crore	~₹10 crore
What we still deliver	Ranges and confidence flags, ranked districts, transparent method	Full decision system	Decision system plus live reliability and grid load

In the worst case the platform degrades gracefully: it publishes wider bands and a longer verification queue rather than false precision.

13. Risk register

Risk	What could happen	Mitigation built into the design
VAHAN access and quality	No bulk API; captcha and terms limits; stock ~25% below state figures	Manual-assisted monthly extraction within terms; reconciliation to state figures; uncertainty bands; request a formal feed
Conflicting official counts	Counts differ by up to 2.8×; loss of trust	Reconciliation waterfall; published scope definitions; offer the schema to BEE
Hidden undercount	Missing chargers make gaps look bigger than they are	Capture–recapture estimate; "what we don't know" layer; ranges below 70% completeness
CPO/OEM reluctance	Competitive concern about sharing locations and uptime	Data contract with benefits; aggregated benchmarks only; public locators within terms
Policy volatility	Subsidy cliffs, scheme changes, draft policies not notified	Policy-shock scenarios; norm versioning; High scenario labelled an upper bound
Low utilisation	1–5% observed vs ~9% break-even: stranded assets	Viability layer; site-specific support sizing; deploy in tranches triggered by utilisation
Grid constraints	Sites that cannot be connected	Substation proxy; DISCOM data request; grid-distance penalty in siting
Land and permissions	Recommended site unusable	Ranked alternates per hex; OMC outlets as default candidates
Standards change	CCS2/LECCS/Type 6 uptake, swap growth, megawatt charging	Compatibility matrix is versioned data; yearly review with ICAT/BIS inputs
Informal sector	Unregistered e-rickshaws, informal charging points	Uplift range; separate light-EV and swap layers
Double counting	Same station in several lists; sanctioned vs installed confusion	ML matching + union-find; separate sanctioned / installed / effective status

Model risk and AI use	Over-confident or opaque outputs	Method Cards, uncertainty bands, published validation error, human sign-off on every Allocation Note, AI-use disclosure
Security and privacy	Cyber risk; obligations under the Digital Personal Data Protection Act, 2023	Aggregated vehicle counts only (no personal data); role-based access; CERT-In empanelled security audit before any government deployment
Build schedule	Six weeks is tight for a student team	Vertical slice in Week 1; MoSCoW scope; documented fallbacks for every Should/Could item

14. Assumptions, limitations and disclosures

- **Pilot numbers** (Odisha and Karnataka) come from our Round 1 pilot run on VAHAN and open charger data; the Odisha official figure of 606 stations is unverified.
- **Costs, tariffs and subsidy intensities** in Section 10 are illustrative planning ranges, to be replaced with tender, quotation and SERC tariff data. The Market/Support/Mandate shares in 10.5 are illustrative; the model estimates them site by site.
- **Low/High light-vs-DC split** in 10.2 uses the Base mix until the model outputs the split directly.
- **Hex-level demand is modelled**, not observed; it carries uncertainty bands and is validated leave-one-district-out.
- **Reliability and source priors** (r_s, τ, P_{work}) are assumptions, listed with ranges in the assumptions register and stress-tested in the tornado chart.
- **Synthetic data**, where used, is flagged per record and listed in the synthetic-data register.
- **AI use:** AI tools assisted with drafting, code and data extraction; the team has reviewed and takes responsibility for every number in this document. AI-extracted rows link to their source pages.